DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING | COURSE PROJECT REVIEW 1

# Automated Retinal Blood Vessel Segmentation Using Mathematical Morphology and Vision Transformers (Morph-ViT)

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|---|---|---|---|
| **Course:** | Fundamentals of Image Processing (FIP) | **Review Date:** | 11/11/2026 (Final Viva: 24/11/2026) |
| **Team Members:** | **Krithin Kusharaj Salian** (NNM24IS110) • **Niriksha** (NNM24IS141) • **Pavan L Poojary** (NNM24IS149) | **Benchmark:** | DRIVE Dataset (40 images, 576×576 resolution) |

## PROJECT ABSTRACT

Automated segmentation of the retinal vascular tree from fundus photography is an essential clinical diagnostic tool for the early detection and management of systemic microvascular diseases, including Diabetic Retinopathy, Arteriosclerosis, Stroke Risk, and Glaucoma. Manual delineation by expert ophthalmologists is time-consuming (1–2 hours per image), prone to inter-observer subjectivity, and impossible to scale for mass screening programs. The segmentation task poses three severe digital image processing challenges: (i) extreme class imbalance (~10% vessel pixels vs. ~90% background), (ii) low local contrast of peripheral microcapillaries, and (iii) non-uniform spherical illumination vignetting.

## PROPOSED METHODOLOGY (MORPH-VIT)

We propose **Morph-ViT**, a hybrid framework that bridges syllabus-grounded mathematical morphology (Units II & III) with Vision Transformer (ViT) self-attention:

1. **Optical Preprocessing & CLAHE:** Hemoglobin absorbs light maximally at 540–580 nm. The green channel is isolated and enhanced using Contrast-Limited Adaptive Histogram Equalization (CLAHE, 8×8 tiles, clip limit 2.0) to normalize background vignetting.
2. **12-Directional Linear Top-Hat Filtering:** To capture vascular structures regardless of orientation, linear structuring elements of length $l = 15$ px are rotated across 12 angles $\theta \in \{0°, 15°, \ldots, 165°\}$. The supremum of white top-hat transformations isolates vascular ridges: $M(x, y) = max_{\theta} [I_{inv} - (I_{inv} \circ L_{\theta})]$.
3. **Deep Hybrid Vision Transformer:** The morphological prior is fused with normalized RGB channels to form a 4-channel tensor. A dual 3×3 CNN stem preserves high-resolution edge geometry, while a 4-layer Vision Transformer encoder (16×16 patches, 4 attention heads) models global vascular connectivity. A convolutional decoder with stem skip connections reconstructs vessel boundaries.
4. **Speck Postprocessing:** Connected-component filtering removes spurious false-positive noise islands ($\leq 10$ px).

## EXPERIMENTAL BENCHMARK RESULTS (DRIVE TEST SET)

All evaluation metrics were calculated **strictly inside the Field of View (FOV)** to prevent artificial inflation from outer border pixels:

| Evaluation Metric | Traditional Method (Top-Hat + CLAHE) | ML/DL Method (Standard U-Net) | Proposed Method (Morph-ViT Hybrid) |
|---|---|---|---|
| **Accuracy** | 89.63% | 88.60% | **92.35% (Best)** |
| **Dice Similarity (F1)** | 64.14% | 72.38% | **77.79% (+13.65% gain)** |
| **IoU (Jaccard Index)** | 47.21% | 56.71% | **63.66% (Best)** |
| **Precision** | 81.19% | 62.81% | **79.09%** |
| **Recall (Sensitivity)** | 53.00% | 85.38% | **76.54% (+23.54% gain)** |
| **Specificity** | 97.40% | 89.28% | **95.71%** |
| **AUC-ROC** | 75.20% | 94.45% | **94.94% (Best)** |
| **Execution Time** | 177.4 ms / img | 671.7 ms / img | **1,099 ms / img (CPU)** |

| Evaluation Metric | Traditional Method (Top-Hat + CLAHE) | ML/DL Method (Standard U-Net) | Proposed Method (Morph-ViT Hybrid) |
|---|---|---|---|
| **Model Size / Parameters** | 0 MB (0 params) | 1.88 MB (482,737 params) | **3.29 MB (862,049 params)** |

## KEY LITERATURE REFERENCES


1. J. Staal et al., “Ridge-based vessel segmentation in color images of the retina,” *IEEE Transactions on Medical Imaging*, vol. 23, no. 4, pp. 501–509, 2004.
2. A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” *ICLR*, 2021.
3. O. Ronneberger et al., “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.
4. J. Serra, *Image Analysis and Mathematical Morphology*, Academic Press, London, 1982.